SENIOR PROJECT

SE Project Hub

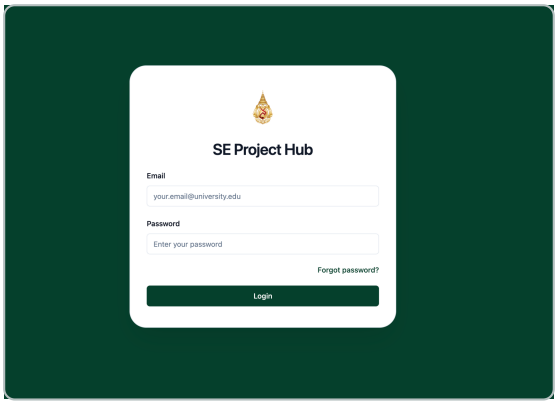
A web platform for managing Software Engineering senior projects from submission to evaluation and archive

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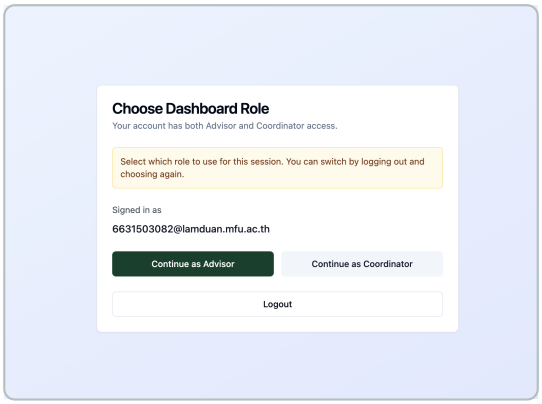
SE Project Hub brings students, advisors, and coordinators into one system. **Coordinators** set up users, courses, and student rosters. **Advisors** follow their courses and evaluate projects against rubrics. **Students** submit their project, track its status, and read feedback. Approved projects end up in a searchable **archive**.

React · TypeScript · Vite · Node.js · Express · Firebase Auth · Supabase PostgreSQL · Supabase Storage · Zod · Helmet · Supertest

Signing In



Login: credentials checked by Firebase Authentication



Choose dashboard: for users who are both Advisor and Coordinator

1. Sign in
The user signs in on the React app through Firebase Authentication.

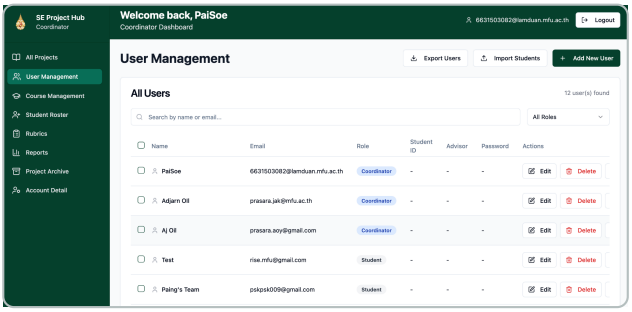
2. Token
The app sends the Firebase ID token with every API request.

3. Verify
Express middleware verifies the token with the Firebase Admin SDK.

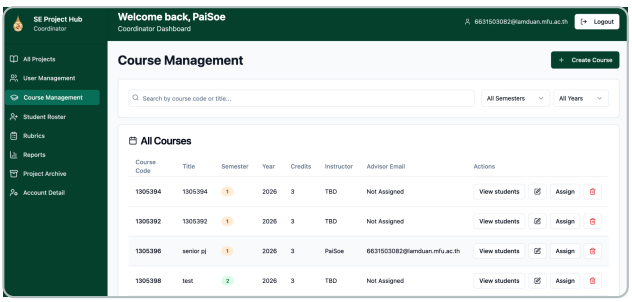
4. Validate
Zod checks the request body before any service code runs.

5. Respond
The service reads or writes Supabase and returns the result.

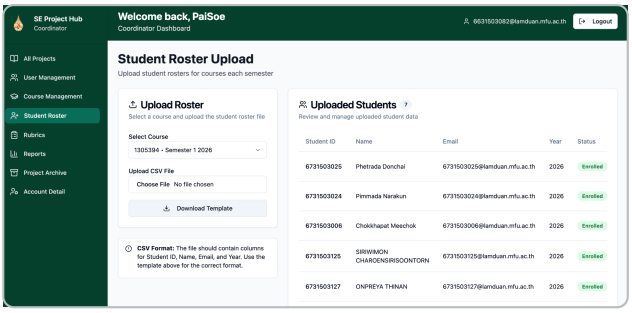
Coordinator



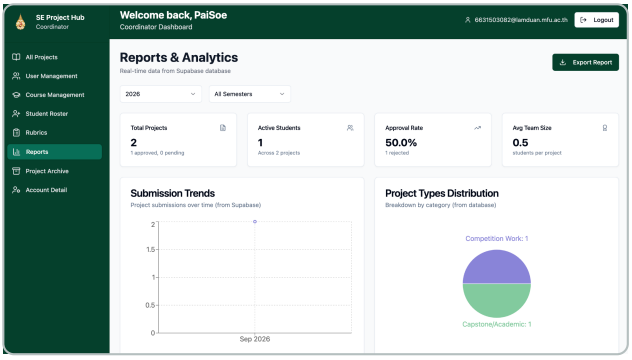
User Management: add, import, edit, export, delete



Course Management: courses, semester, advisors



Roster Upload: enrol students from a CSV



Reports and Analytics: project statistics

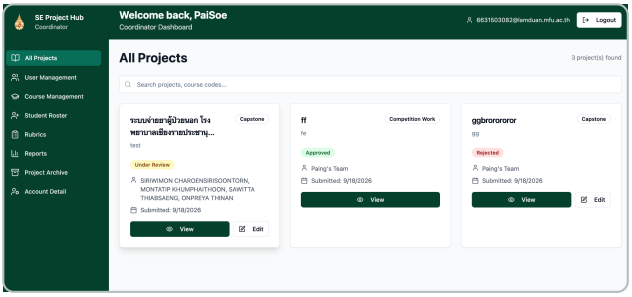
Set up a semester

- **Users**: search, filter, add, import, edit, export, and delete user records.
- **Courses**: code, title, semester, academic year, instructor, and advisor assignment.
- **Roster**: upload a CSV for a course, then review the imported students.

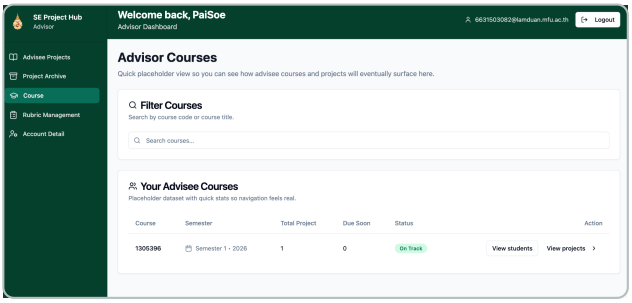
Reports

- Filter by **academic year** and **semester**.
- Project counts, active students, approval rates, and average team size.
- Submission trends and project-type distribution.

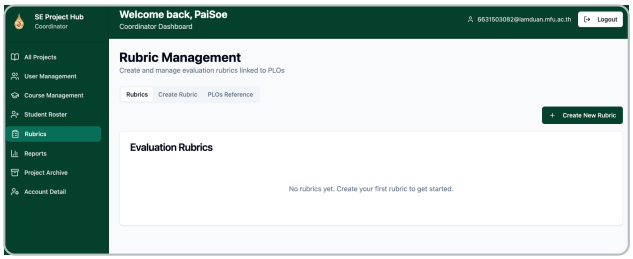
Advisor



All Projects: search and open submissions



Advisor Courses: assigned courses and students



Rubric Management: criteria linked to PLOs

Projects

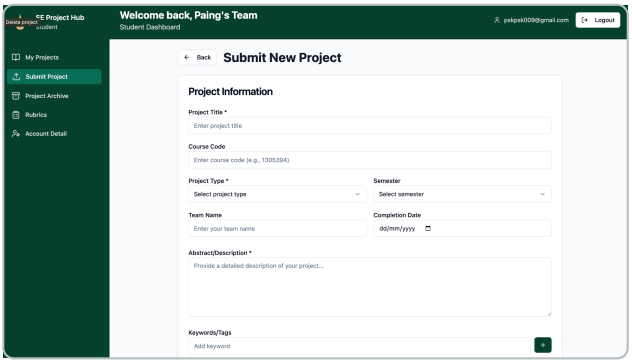
- **All Projects** lists name, type, status, team, and submission date, for Advisors and Coordinators.
- **Advisor Courses** shows each assigned course with its enrolled students and their projects.
- Editing depends on the user's role and the project's status.

Evaluation

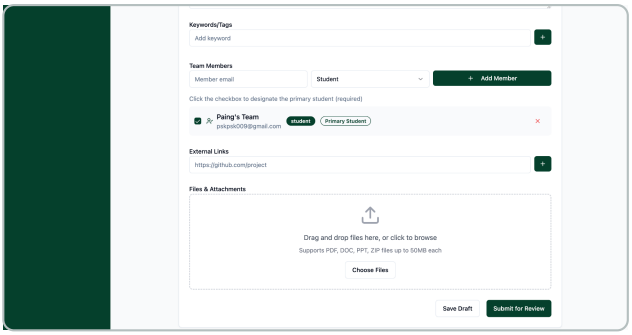
- A rubric is made of **criteria** and **performance levels**.
- Criteria are linked to **Program Learning Outcomes**.
- Advisors score a project against the rubric and save the scores with feedback.

Student

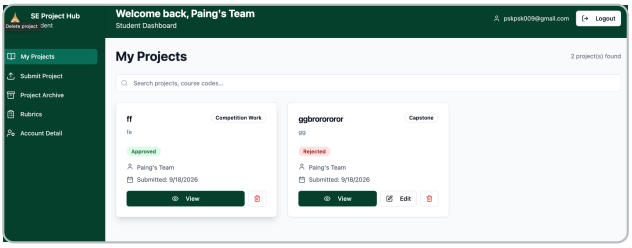
Students do everything from their own dashboard: submit the team's project, follow it through review, and check the rubric it will be graded on.



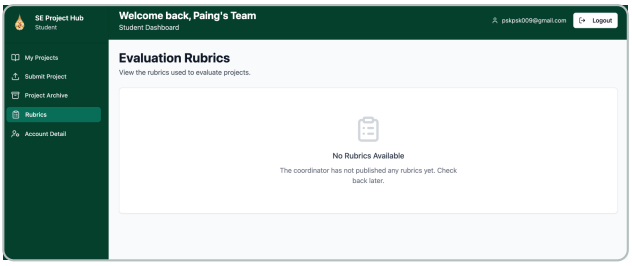
Submit Project (1): name, course, type, team



Submit Project (2): links, files, submit



My Projects: status and feedback



Student Rubrics: how projects are assessed

1. Fill in
Project name, course, type, and team members.

2. Attach
Links and supporting files, stored in Supabase Storage.

3. Submit
The project is saved with its team in `team_member`.

4. Review
The advisor evaluates it with the rubric and leaves feedback.

5. Track
The student sees the status and feedback, and resubmits if allowed.

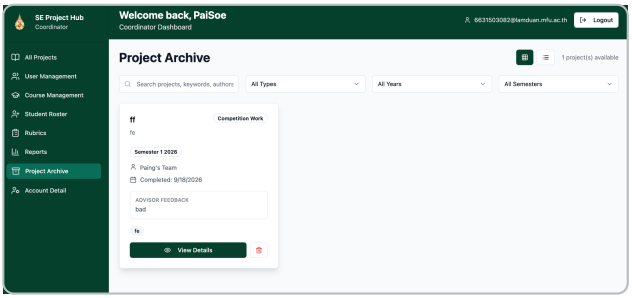
Submitting

- The form takes the **project name**, **course**, **project type**, and **team members**.
- Students can add **links** and upload **supporting files**.
- The second view lets them review everything before they submit.

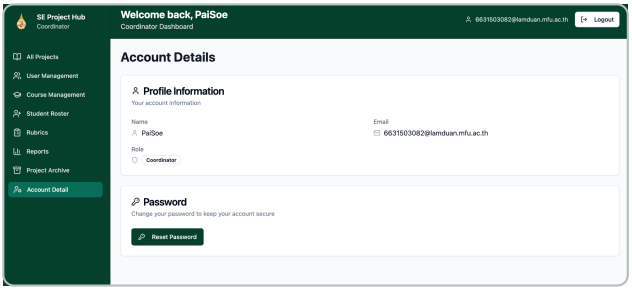
After submitting

- **My Projects** lists each submission with its current **status**.
- Students open a project to read the advisor's **feedback**, and can edit or resubmit when the workflow allows it.
- **Student Rubrics** shows the criteria and performance levels in read-only form, so students know how they will be assessed.

Every User



Project Archive: search past projects



Account Details: profile information

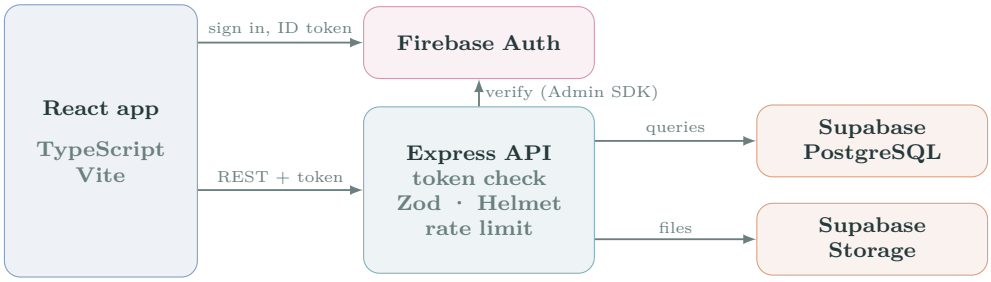
Archive

- Approved or completed projects stay searchable after the semester.
- Search by keyword, project type, academic year, and semester.

Account

- Shows the user's profile and roles.
- Account options supported by the application.

Architecture & Data Model



Main data

- **Users** with Student, Advisor, and Coordinator roles
- **Courses** and their student rosters
- **Projects** ↔ students through `team_member`
- **Rubrics** → criteria → performance levels, plus evaluation records

Reliability

- Every route validates its input with **Zod**.
- **Helmet** security headers and per-IP **rate limiting**.
- Central error handling that never exposes internals.
- Unit tests (CSV roster parsing) and **Supertest** route tests.